

Entity interpolation for high speed projectiles is wildly inaccurate

[MC-299627](#) NETWORKING

Status	Resolved / Fixed
Affected	1.21.7
Fixed in	25w32a
Reported	2025-07-10
Mojira category	Networking; Projectiles

REPRODUCTION TARGET

Use version:

1.21.7

Affected versions (1):

1.21.7

Environment:

Minecraft Java Edition 1.21.7 in a single-player Creative-mode world with command permissions available. Use an open area with an unobstructed view of the sky. Vanilla entity hitboxes must be visible before the summon command is executed. No particular seed, coordinates, graphics setting, resource pack, mod, or dedicated server is specified by the report.

PRECONDITIONS

- Launch exactly Minecraft Java Edition 1.21.7; do not use fix version 25w32a for this reproduction.
- Use a world in which the player can execute commands. Creating a new Creative-mode single-player world is the simplest implementation because it provides command access and easy movement.
- Ensure the player is in an open location where blocks, trees, cave ceilings, or structures do not obstruct the sky directly above the summon position.
- Use the fourth command supplied by the report, because Comment 2 explicitly states that executing the fourth command while looking upward with hitboxes enabled is sufficient.
- Be ready to begin observing immediately when the command is submitted, because the summoned arrow starts moving as soon as it is created.

STEPS TO REPRODUCE

1 **REPORT**

From the Minecraft Launcher, start a Java Edition installation configured for version 1.21.7. Wait at the title screen until the version identifier confirms that 1.21.7 is running.

SUCCESS CUE The Minecraft Java Edition title screen is loaded and displays version 1.21.7.

2 **INFERRED**

Select Singleplayer, choose Create New World, set the game mode to Creative, and create the world. If using an existing world instead, ensure commands are permitted before continuing.

SUCCESS CUE The player is fully loaded into a world and has Creative-mode command access.

3 **INFERRED**

Wait for the nearby terrain to finish rendering. Confirm that looking around and moving responds normally before preparing the command.

SUCCESS CUE Surrounding chunks are rendered and player controls respond without world-loading interruption.

4 **INFERRED**

Position the player beneath an unobstructed section of sky. If necessary in Creative mode, double-tap the Jump control, which is Space by default, to begin flying and move away from trees, overhangs, cave ceilings, or structures. Stop in a stable position with clear vertical space above the player.

INPUT Space twice, if Creative flight is needed

SUCCESS CUE Looking directly upward from the player's position shows open sky rather than a nearby block or structure.

5 **REPORT**

Enable Minecraft's entity hitbox display by pressing the debug hitbox shortcut. With default controls this is F3+B; on keyboards where function keys require a modifier, hold Fn as well. Release the keys after toggling the display on.

INPUT F3+B

SUCCESS CUE A debug confirmation indicates that hitboxes are shown; visible nearby entities, if any, have hitbox outlines.

6 **REPORT**

Aim the camera upward into the unobstructed sky, preferably nearly straight up. Keep the camera in this orientation so that the high-speed arrow's path begins within the visible part of the screen.

SUCCESS CUE The crosshair points upward into open sky and no block is directly between the player and the sky.

7 **INFERRED**

Press the Open Chat control, which is T by default. Do not turn the camera after opening chat.

INPUT T

SUCCESS CUE The chat command input is open while the view remains aimed upward.

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REPORT

Enter the fourth summon command from the report exactly as written, including the leading slash, capitalization, coordinates, NBT braces, decimal values, commas, and brackets. Do not submit it yet until observation can begin immediately.

INPUT `/summon minecraft:arrow ~ ~1 ~ {Motion:[1.0,10.0,0.0]}`

SUCCESS CUE The complete command `/summon minecraft:arrow ~ ~1 ~ {Motion:[1.0,10.0,0.0]}` is present in the chat input with no alterations.

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REPORT

TRIGGER

Press Enter once to execute the command. Keep looking upward and do not immediately move the player, turn the camera, disable hitboxes, pause, or open another interface.

INPUT `Enter`

SUCCESS CUE The command is accepted without a red syntax or permission error, chat closes, and an arrow has been summoned one block above the player with the supplied motion while hitbox rendering remains enabled.

HANDOFF TO OBSERVATION

The command `/summon minecraft:arrow ~ ~1 ~ {Motion:[1.0,10.0,0.0]}` has just been executed once in Minecraft Java Edition 1.21.7 while the camera is aimed upward into open sky and entity hitboxes are enabled. The arrow is now in flight; hand off immediately to the observation subsystem without changing the camera or debug-display state.

UNCERTAINTIES

- The description lists four arrow summon commands, but Comment 2 states that the fourth command alone is sufficient while looking upward with hitboxes enabled. This plan therefore uses only the fourth command as the shortest deterministic trigger.
- Single-player Creative mode and the use of an open-sky test location are practical execution choices. The report requires command execution and an upward view but does not mandate a particular game mode, seed, coordinate, or world type.
- The report does not explain how the attachment's colored green and white visualization was produced. This plan uses Minecraft's standard F3+B hitbox display because Comment 2 explicitly confirms that hitboxes are sufficient, and it does not attempt to recreate any unspecified attachment-only overlay.
- The formal Affects Versions field lists only 1.21.7. Although a later comment claims the problem also remains in 1.21.8, 1.21.8 is not added to `affected_versions` because it is not present in the report's formal affected-version list.

MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Trigger reached?	YES / NO
Crash file	
Notes	
Tested by / date	